Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The prefix for the fraction 10^{-15} is

(A) nano

(B) pico

(C) femto

(D) giga

Q.2

Which of the following expressions is not dimensionally correct?

(A) $\frac{T}{\mathrm{K}} = \frac{t_{\mathrm{c}}}{^{\circ}\mathrm{C}} + 273.15$

(B) $\mathrm{pH} = -\log\left[\mathrm{H}^{+}\right]$

(C) $\frac{T}{\mathrm{K}} = \left(\frac{t_{\mathrm{F}}}{^{\circ}\mathrm{F}} - 32\right)\left(\frac{5}{9}\right) + 273.15$

(D) $\frac{T_{\mathrm{F}}}{^{\circ}\mathrm{F}} = \left(\frac{9}{5}\right)\left(\frac{t_{\mathrm{c}}}{^{\circ}\mathrm{C}}\right) + 32$

Q.3

The number of significant numbers in 0.300 are

(A) 2

(B) 3

(C) 1

(D) 4

Q.4

The radius of a hydrogen atom is $5.29 \times 10^{-11}\ \mathrm{m}$ and that of a proton is $1.5 \times 10^{-15}\ \mathrm{m}$. The ratio of the radius of atom to the radius of proton in the scientific notation will be

(A) 3.526×10^{4}

(B) 35.266×10^{3}

(C) 3.5×10^{4}

(D) 3.5266667×10^{3}

Q.5

If $\pi = 3.14159$, then the value of $5\pi(0.0125)$ in the scientific notation will be

(A) 0.001964

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(B) 1.96×10^{-3}

(C) 1.9635×10^{-3}

(D) 3.5266667×10^{-2}

Q.6

The answer to the expression $1.561 \times 10^{3} - 1.80 \times 10^{2} + 2.02 \times 10^{4}$ upto the correct significant digits is

(A) 2.16×10^{4}

(B) 2.158×10^{4}

(C) 2.1581×10^{4}

(D) 21.59×10^{4}

Q.7

The number of significant digits in the number 126000 is

(A) 3

(B) 4

(C) 5

(D) 6

Q.8

Chlorine exists as $^{35}\mathrm{Cl}$ (atomic mass $= 34.9688\mathrm{u}; 75.77\%$) and $^{37}\mathrm{Cl}$ (atomic mass $= 36.9659\mathrm{u}; 24.23\%$). The atomic mass of natural existing chlorine is

(A) $35.4527\mathrm{u}$

(B) $36.4236\mathrm{u}$

(C) $34.1235\mathrm{u}$

(D) $32.3458\mathrm{u}$

Q.9

An organic compound contains $20.0\%\mathrm{C}, 6.66\%\mathrm{H}, 47.33\%$ N and the rest was oxygen. Its molar mass is 60 g mol^{-1}. The molecular formula of the compound is

(A) $\mathrm{CH_4\ N_2O}$

(B) $\mathrm{CH_2NO}$

(C) $\mathrm{C_2H_6NO}$

(D) $\mathrm{CH_{18}NO}$

Q.10

If an iron oxide has 69.9 mass $\%$ of Fe (molar mass $= 56.0$ g mol^{-1}) and 30.1 mass $\%$ of oxygen, its molecular formula will be

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(A) FeO

(B) Fe_2O_3

(C) Fe_3O_4

(D) Fe_2O_6

Q.11

The number of oxygen atoms in $24.9\ g$ of $CuSO_4 \cdot 5H_2O$ is (molar mass of $Cu = 63\ g\ mol^{-1}$)

(A) 2.41×10^{24}

(B) 3.01×10^{24}

(C) 5.42×10^{23}

(D) 5.42×10^{24}

Q.12

A compound contains $11.99\%\ N, 13.70\%O, 9.25\%\ B$ and $65.06\%\ F$. Its empirical formula is (molar mass of B is $10.8\ g\ mol^{-1}$)

(A) $NOBF_2$

(B) $NOBF_4$

(C) N_2OF_2

(D) $NO_2\ F_2$

Q.13

In the reaction $N_2H_4 + 3O_2 \rightarrow 2NO_2 + 2H_2O$, the mass of O_2 required to combine with $745\ g$ of N_2H_4 will be

(A) $2120\ g$

(B) $2235\ g$

(C) $2436\ g$

(D) $2510\ g$

Q.14

A mixture of $50.0\ g$ of S and $100.0\ gCl_2$ reacts of form S_2Cl_2. The mass of S_2Cl_2 formed will be

(A) $150.0\ g$

(B) $105.5\ g$

(C) $121.0\ g$

(D) $135.1\ g$

Q.15

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1.8 g of Mg is burnt in a closed vessel which contains 0.8 g of oxygen. Which of the following statements is not correct? Given ; $M(\mathrm{Mg}) = 24.3\ \mathrm{g\ mol^{-1}}$.

(A) 0.05 mol of MgO is formed

(B) 0.8 g of Mg is left behind

(C) Oxygen is completely used in the reaction

(D) The extent of reaction at the completion of reaction is 0.025 mol.

Q.16

The mass of ethanol (molar mass = $46\ \mathrm{g\ mol^{-1}}$) to be added to 1.0 kg of water so as to have its amount fraction equal to 0.2 is

(A) 319.5 g

(B) 432.1 g

(C) 638.9 g

(D) 719.3 g

Q.17

A solution of acetic acid has molarity equal to $1.35\mathrm{M}$ and molality equal to $1.45\ \mathrm{mol\ kg^{-1}}$. The density of solution will be

(A) $1.251\ \mathrm{g\ mL^{-1}}$

(B) $1.125\ \mathrm{g\ mL^{-1}}$

(C) $1.012\ \mathrm{g\ mL^{-1}}$

(D) $0.994\ \mathrm{g\ mL^{-1}}$

Q.18

The density of a 3 molal solution of sodium thiosulphate $(\mathrm{Na_2\ S_2O_3 \cdot 5H_2O})$ solution is $1.25\ \mathrm{g\ cm^{-3}}$. The percentage by mass of sodium thiosulphate is

(A) 35.25%

(B) 39.20%

(C) 42.67%

(D) 46.67%

Q.19

The molarity of a sulphuric acid solution is $2.32\ \mathrm{moldm^{-3}}$. If the density of solution is $1.14\ \mathrm{g\ cm^{-3}}$, the molality of the solution will be

(A) $2.54\ \mathrm{mol\ kg^{-1}}$

(B) $2.25\ \mathrm{mol\ kg^{-1}}$

(C) $2.62\ \mathrm{mol\ kg^{-1}}$

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(D) 1.98 mol kg^{-1}

Q.20

If 20.0 cm^3 of $1.0 \text{M} \text{CaCl}_2$ and 60.0 cm^3 of $0.20 \text{M} \text{CaCl}_2$ are mixed, the molarity of the resultant solution is

(A) 0.80M

(B) 0.60M

(C) 0.40M

(D) 0.20M

Q.21

Boron occurs in two varieties, namely, ^{10}B (atomic mass: 10.01amu) and ^{11}B (atomic mass: 11.01amu. The atomic mass of naturally occurring element is reported as 10.82amu. The per cent of ^{10}B in this naturally occurring boron is

(A) 10

(B) 19

(C) 29

(D) 35

Q.22

Which of the following physical quantities has SI unit of $\text{kgm}^2 \text{ s}^{-3} \text{ A}^{-2}$?

(A) Electric resistance

(B) Electric potential

(C) Power

(D) Energy

Q.23

The molarity of concentrated sulphuric acid $\left(\rho = 1.834 \text{ g cm}^{-3}\right)$ containing 95% of H_2SO_4 by mass is

(A) 4.44M

(B) 8.88M

(C) 13.32M

(D) 17.78M

Q.24

The volume V of $\text{K}_2\text{Cr}_2\text{O}_7$ (molar mass 294 g mol^{-1}) solution (mass concentration 0.005 g per mL) is equivalent to 35.0 mL of $0.02 \text{M} \text{KMnO}_4$ (molar mass 158 g mol^{-1}), when these solutions are used in the titration in acid solution. The value of V would be

(A) 46.75 mL

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(B) $34.3\ mL$

(C) $29.05\ mL$

(D) $22.5\ mL$

Q.25

The mass of Na_2CO_3 (91.2% purity) required for the neutralisation of $45.6\ mL$ of $0.25M\ HC1$ solution is

(A) $0.6625\ g$

(B) $0.4265\ g$

(C) $0.5765\ g$

(D) $0.8473\ g$

Answers & Solutions

Q.1 Answer:

femto

Solution:

nano	n	10^{-9}
pico	p	10^{-12}
femto	**f**	10^{-15}
atto	a	10^{-18}

Q.2 Answer:

$pH = -\log[H^+]$

Solution:

$pH = -\log[H^+]$, . . . **Because pH is a measure of hydrogen ion concentration, it is used to quantitatively characterize solutions as acidic, neutral, or basic (alkaline). The typical pH scale runs from 0 - 14**

Q.3 Answer:

3

Solution:

Zeroes at the end or on the right side of the number are also significant, Therefore, 0.300 has three significant figures.

Q.4 Answer:

3.5×10^4

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Solution:

$$\frac{5.29 \times 10^{-11}\ \mathrm{m}}{1.5 \times 10^{-15}\ \mathrm{m}} = 3.5 \times 10^{4}$$

(Result will carry 2 significant digits.)

Q.5 Answer:

1.96×10^{-3}

Solution:

Result will carry 3 significant digits.

Q.6 Answer:

2.16×10^{4}

Solution:

The result will carry 2 digits after decimal.

Q.7 Answer:

3

Solution:

There are 3 digits. The last three zeros are not significant.
There are 3 significant digits because all non-zero digits are in a number are significant and the terminal zeros are not significant if there is no decimal point.

Q.8 Answer:

$35.4527\mathrm{u}$

Solution:

$m_{\mathrm{a}} = (0.7577 \times 34.9688 + 0.2423 \times 36.9659)\mathrm{u} = 35.4527\mathrm{u}$

Q.9 Answer:

$\mathrm{CH_4}$ $\mathrm{N_2O}$

Solution:

$$\mathrm{C : N : O : H} :: \frac{20}{12} : \frac{47.33}{14} : \frac{26.01}{16} : \frac{6.66}{1} :: 1.667 : 3.381 : 1.626 : 6.66 :: 1 : 2 : 1 : 4$$

Empirical formula $\mathrm{CN_2OH_4}$
Molar empirical mass is $60\ \mathrm{g\ mol^{-1}}$ (same as the given molar mass). Hence, Molecular formula is $\mathrm{CN_2OH_4}$.

Q.10 Answer:

$\mathrm{Fe_2O_3}$

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Solution:

The iron oxide has 69.9% iron and 30.1% dioxygen by mass.
Thus, 100 g of iron oxide contains 69.9 g iron and 30.1 g dioxygen.
The number of moles of iron present in 100 g of iron oxide are $\frac{69.9}{55.8} = 1.25$.
The number of moles of dioxygen present in 100 g of iron oxide are $\frac{30.1}{32} = 0.94$.
The ratio of the number of oxygen atoms to the number of iron atoms present in one formula unit of iron oxide is $\frac{2\times0.94}{1.25} = 1.5 : 1 = 3 : 2$.
Hence, the formula of the iron oxide is Fe_2O_3.

Q.11 Answer:

5.42×10^{23}

Solution:

Molar mass of $CuSO_4 \cdot 5H_2O = (63 + 32 + 4 \times 16 + 5 \times 18)\text{gmol}^{-1} = 249\text{ g mol}^{-1}$ Amount of $CuSO_4 \cdot 5H_2O$ in the given mass is $n = \frac{m}{M} = \frac{24.9\text{ g}}{249\text{ g mol}^{-1}} = 0.1\text{ mol}$ Number of oxygen atoms $= 9 \times (0.1\text{ mol})\left(6.022 \times 10^{23}\text{ mol}^{-1}\right) = 5.42 \times 10^{23}$

Q.12 Answer:

$NOBF_4$

Solution:

$$N : O : B : F :: \frac{11.99}{14} : \frac{13.7}{16} : \frac{9.25}{10.8} : \frac{65.06}{19} :: 0.856 : 0.856 : 0.856 : 3.424 :: 1 : 1 : 1 : 4$$

Empirical formula: $NOBF_4$

Q.13 Answer:

2235 g

Solution:

We have

$$\underset{32\text{ g}}{N_2H_4} + \underset{}{3O_2} \rightarrow \underset{}{2NO_2} + \underset{96\text{ g}}{2H_2O}$$

Mass of O_2 required $= \left(\frac{96\text{ g}}{32\text{ g}}\right)(745\text{ g}) = 2235\text{ g}$

Q.14 Answer:

105.5 g

Solution:

The reaction is

$$\underset{n_0}{2S} + \underset{50.0\text{ g}}{Cl_2} \rightarrow \underset{100.0\text{ g}}{S_2Cl_2}$$

$n_0(\text{ S}) = \frac{50.0\text{ g}}{32\text{ g mol}^{-1}} = 1.5625\text{ mol}$
$n_0\,(Cl_2) = \frac{100.0\text{ g}}{71.0\text{ g mol}^{-1}} = 1.4084\text{ mol}$

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$\xi_{\max}(\mathrm{S}) = \frac{n_0(\mathrm{S})}{v(\mathrm{S})} = \frac{1.5625 \text{ mol}}{2} = 0.7812 \text{ mol}$

$\xi_{\max}(\mathrm{Cl_2}) = \frac{n_0(\mathrm{Cl_2})}{v(\mathrm{Cl_2})} = \frac{1.4084 \text{ mol}}{1} = 1.4084 \text{ mol}$

S acts as limiting reagent. Hence

$\Delta n(\mathrm{S_2Cl_2}) = v(\mathrm{S_2Cl_2})\,\xi_{\max}(\mathrm{S}) = (1)(0.7812 \text{ mol}) = 0.7812 \text{ mol}$

$\Delta m(\mathrm{S_2Cl_2}) = \Delta n(\mathrm{S_2Cl_2})\,M(\mathrm{S_2Cl_2}) = (0.7812 \text{ mol})\left(135 \text{ g mol}^{-1}\right) = 105.5 \text{ g}$

Q.15 Answer:

0.8 g of Mg is left behind

Solution:

The reaction is $2\mathrm{Mg} + \mathrm{O_2} \rightarrow 2\mathrm{MgO}$

$\xi_{\max}(\mathrm{Mg}) = \frac{m_{\mathrm{Mg}}/M_{\mathrm{Mg}}}{v_{\mathrm{Mg}}} = \frac{1.8 \text{ g}/24.3 \text{ g mol}^{-1}}{2} = 0.037 \text{ mol}$

$\xi_{\max}(\mathrm{O_2}) = \frac{0.8 \text{ g}/32 \text{ g mol}^{-1}}{1} = 0.025 \text{ mol}$

Since $\xi_{\max}(\mathrm{O_2}) < \xi_{\max}(\mathrm{Mg})$**, the limiting reagent is oxygen.**

$$\Delta n_{\mathrm{MgO}} = v_{\mathrm{MgO}}\xi_{\max}(\mathrm{O_2}) = 2(0.025 \text{ mol}) = 0.05 \text{ mol}$$

Mass of Mg **consumed** $= \left(\frac{2M_{\mathrm{Mg}}}{M_{\mathrm{O_2}}}\right) m_{\mathrm{O_2}} = \frac{48.6}{32} \times 0.8 \text{ g} = 1.22 \text{ g}$

Mass of Mg **left behind** $= 1.8 \text{ g} - 1.22 \text{ g} = 0.58 \text{ g}$

Q.16 Answer:

638.9 g

Solution:

By definition, $x_2 = n_2/(n_1 + n_2)$

Hence, $x_2(n_1 + n_2) = n_2$ **or** $x_2 n_1 = (1 - x_2)\,n_2$

or

$x_2 n_1 = x_1 n_2$ **or** $n_2 = x_2 n_1 / x_1$

Now, $n_1 = \frac{m_1}{M_1} = \frac{1000 \text{ g}}{18 \text{ g mol}^{-1}} = 55.56 \text{ mol}$

Hence, $n_2 = (0.2)(55.56 \text{ mol})/(0.8) = 13.89 \text{ mol}$

Q.17 Answer:

1.012 g mL^{-1}

Solution:

For $1.35\mathrm{M}$ **solution, we will have** $n_2 = 1.35 \text{ mol}$ **and** $V = 1000 \text{ mL}$ **The solution is also** 1.45 mol kg^{-1}**. The mass of solvent to have** 1.35 mol **of solute will be**

$m_1 = \left(\frac{1.35 \text{ mol}}{1.45 \text{ mol kg}^{-1}}\right) = 0.93103 \text{ kg} = 931.03 \text{ g}$

The mass of solution will be $m = m_1 + n_2 M_2 = 931.03 \text{ g} + (1.35 \text{ mol})\left(60 \text{ g mol}^{-1}\right) = 1012.03 \text{ g}$

The density of the solution will be $\rho = \frac{m}{V} = \frac{1012.03 \text{ g}}{1000 \text{ mL}} = 1.012 \text{ g mL}^{-1}$

Q.18 Answer:

42.67%

Solution:

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Molar mass of $Na_2S_2O_3 \cdot 5H_2O$ is $M = (2 \times 23 + 2 \times 32 + 3 \times 16 + 5 \times 18) \text{gmol}^{-1} = 248 \text{ g mol}^{-1}$ **The mass of 3 mol of $Na_2S_2O_3 \cdot 5H_2O$ is** $m_2 = (3 \text{ mol})\left(248 \text{ g mol}^{-1}\right) = 744 \text{ g}$
The mass of solution will be $m = m_1 + m_2 = 1000 \text{ g} + 744 \text{ g} = 1744 \text{ g}$
Mass percentage of sodium thiosulphate $= \frac{m_2}{m} \times 100 = \left(\frac{744 \text{ g}}{1744 \text{ g}}\right) \times 100 = 42.67\%$

Q.19 Answer:

2.54 mol kg^{-1}

Solution:

For 1 L of solution, we have $n_2 = 2.32 \text{ mol}$ and $V = 1000 \text{ cm}^3$
Mass of solution, $m = V\rho = \left(1000 \text{ cm}^3\right)\left(1.14 \text{ g cm}^{-3}\right) = 1140 \text{ g}$
Mass of sulphuric acid, $m_2 = n_2 M_2 = (2.32 \text{ mol})\left(98 \text{ g mol}^{-1}\right) = 227.36 \text{ g}$
Mass of solvent, $m_1 = m - m_2 = 1140 \text{ g} - 227.36 \text{ g} = 912.64 \text{ g} = 0.913 \text{ kg}$
Molality of solution $= \frac{n_2}{m_1} = \frac{2.32 \text{ mol}}{0.913 \text{ kg}} = 2.54 \text{ mol kg}^{-1}$

Q.20 Answer:

0.40M

Solution:

The molarity of the resultant solution will be

$$M = \frac{M_1V_1 + M_2V_2}{V_1 + V_2} = \frac{(1.0\text{M})\left(20 \text{ cm}^3\right) + (0.20\text{M})\left(60.0 \text{ cm}^3\right)}{(20 \text{ cm}^3 + 60 \text{ cm}^3)} = \frac{20 + 12}{80}\text{M} = 0.40\text{M}$$

Q.21 Answer:

19

Solution:

If x is the fraction of ^{10}B, we have $x(10.01\text{amu}) + (1 - x)(11.01\text{amu}) = 10.82\text{amu}$ **This gives $x = 0.19$. Hence, percentage is 19%.**

Q.22 Answer:

Electric resistance

Solution:

By definition, $I = E/R$. Hence,

$$\text{Unit of resistance } = \frac{\text{Unit of potential}}{\text{Unit of current}} = \frac{\text{kgm}^2 \text{ s}^{-3} \text{ A}^{-1}}{\text{A}} = \text{kgm}^2 \text{ s}^{-3} \text{ A}^{-2}$$

Q.23 Answer:

17.78M

Solution:

Molarity $= \frac{95 \text{ g}/98 \text{ g mol}^{-1}}{100 \text{ g}/1.834 \text{ g cm}^{-3}} = 0.01778 \text{ mol cm}^{-3} = 17.78 \text{ moldm}^{-3}$

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Q.24 Answer:

$34.3\ \text{mL}$

Solution:

The reactions are

$$\frac{1}{6}\text{Cr}_2\text{O}_7^{2-} + \frac{14}{6}\text{H}^+ + \text{e}^- \longrightarrow \frac{1}{3}\text{Cr}^{3+} + \frac{7}{6}\text{H}_2\text{O}$$

and

$\frac{1}{5}\text{MnO}_4^- + \frac{8}{5}\text{H}^+ + \text{e}^- \longrightarrow \frac{1}{5}\text{Mn}^{2+} + \frac{4}{5}\text{H}_2\text{O}$

Thus, $n\left(\frac{1}{6}\text{Cr}_2\text{O}_7^{2-}\right) \equiv n\left(\frac{1}{5}\text{MnO}_4^-\right)$ **i.e.** $6n\left(\text{Cr}_2\text{O}_7^{2-}\right) = 5n\left(\text{MnO}_4^-\right)$

Now

$$\begin{aligned}
&\text{Now} \quad && n\left(\text{KMnO}_4\right) = VM = \left(0.035\ \text{L}\right)\left(0.02\ \text{mol L}^{-1}\right) = 0.0007\ \text{mol} \\
& && n\left(\ \text{K}_2\text{CrO}_7\right) = VM = V\left(\frac{0.005\ \text{g mL}^{-1}}{294\ \text{g mol}^{-1}}\right) = V\left(\frac{0.005}{294}\ \text{mol mL}^{-1}\right) \\
&\text{Thus} \quad && 6V\left(\frac{0.005}{294}\ \text{mol mL}^{-1}\right) = 5(0.0007\ \text{mol}) \quad \text{or} \quad V = \left(\frac{5}{6}\right)\left(\frac{0.0007\times 294}{0.005}\right)\text{mL} = 34.3\ \text{mL}
\end{aligned}$$

Q.25 Answer:

$0.6625\ \text{g}$

Solution:

The reaction is $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{CO}_3$

Amount of HCl **to be neutralised** $= V \times M = \left(45.6 \times 10^{-3}\ \text{L}\right)\left(0.25\ \text{mol L}^{-1}\right) = 11.4 \times 10^{-3}\ \text{mol}$

Amount of Na_2CO_3 **required** $= 0.5 \times 11.4 \times 10^{-3}\ \text{mol} = 5.7 \times 10^{-3}\ \text{mol}$

Mass of Na_2CO_3 **required** $= \left(5.7 \times 10^{-3}\ \text{mol}\right)\left(106\ \text{g mol}^{-1}\right)(100/91.2) = 0.6625\ \text{g}$

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